

Problem B: Book Recommendation

The development of information technology and the internet has ended an information-shortage era, but has brought the public overloaded information. In the present era, both consumers and producers of information are facing significant challenges: consumers find it is harder to have access to information they are interested in, with explosively growing information around them; and producers also find it is harder to make their information focused by readers.

The conflicts could be solved by a tool called ‘Recommendation’. Recommendations are widely used in network products and Apps. Typical styles include the relevant search, topics recommendations, products push in the e-commercial, and ‘recommended friends’ search in various SNSs.

Attached data are the information of users’ behavior at a popular on-line bookstore, including remarks on books, tags of books, and the users’ social networks. Consider the following question with the attached data:

1. Analyze the effects on book remarks by users.
2. Design a model to predict the book remarks by users on their unread books. Information of users is attached in the file ‘predict.txt’.
3. Recommend every user (attached in the file ‘predict.txt’) three books, which they have yet read before.

Data instruction

The file 'predict.txt' includes the set of users for your prediction, with two columns. The left column is the users' ID, and the right is the book's ID. The corresponding predicted remarks should be listed in the third column.

The file 'user_book_score.txt' indicates the data of users' remarks on books, with three columns. The columns are users' ID, the books' ID, and the remarks on books, from left to right. (Remarks range from 1 to 5. The higher the score of the book is, the more endorsement of users on the book.)

The file 'book_tag.txt' indicates the tags of books, with two columns. The left column is the books' ID, and the right is the tag of the corresponding book. Tags are separated by semicolons. The value of each tag varies with the popularity of the book.

The file 'user_social.txt' indicates the social networks of users, with two columns. The left column is the users' ID, and the right is the follows of the corresponding user. In this file, the relationships are unidirectional.

The file 'user_read_history.txt' indicates the users' reading history, with two columns. The left column is the users' ID, and the right is the ever read books of the corresponding user.